LESSON 6.12

Least Common Multiple

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the least common multiple (LCM) of two numbers by listing or comparing their multiples.

WHY THIS MATTERS Astronaut Reyes checks the oxygen system every 4 hours. Astronaut Chen checks the water recycler every 6 hours. They both start their shifts at 6:00 AM. When is the next time both astronauts will be checking their systems at the same time?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Least Common Multiple *Spanish: Mínimo común múltiplo*	The least positive number that is a multiple of two or more numbers.	Multiples of 4 and 6 first meet at 12, so LCM = 12.
Multiple *Spanish: Múltiplo*	What you get when you multiply a number by 1, 2, 3, and so on.	Multiples of 4: 4, 8, 12, 16, 20 — found by multiplying 4 × 1, 4 × 2, 4 × 3, ...
Common multiple *Spanish: Múltiplo común*	A number that two or more numbers both go into.	12, 24, and 36 are all common multiples of both 3 and 4
Skip counting *Spanish: Conteo salteado*	Counting by a number, like 2, 4, 6, to list its multiples.	Skip count by 6: 6, 12, 18, 24, 30 — each jump adds 6
Prime factorization *Spanish: Factorización prima*	Writing a number as prime numbers multiplied together. It helps you find the LCM.	4 = 2 × 2, 6 = 2 × 3. LCM uses the highest power of each prime: $2^2 \times 3 = 12$

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

I want the LCM of 4 and 6. I skip count the multiples of 4: 4, 8, 12, 16, 20, 24.

1. Then I skip count the multiples of 6: 6, 12, 18, 24.
2. I look for the first number in BOTH lists. 12 is in both lists, and it comes before 24.
3. So the LCM of 4 and 6 is 12. I check: 12 ÷ 4 = 3 and 12 ÷ 6 = 2, so both go into 12 evenly.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's find the LCM of 3 and 5. What are the multiples of 3? (3, 6, 9, 12, 15.)

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The LCM of 4 and 6 is _____ because it is the smallest number in the _____ group.

Word bank Least Common Multiple | Multiple | Common multiple | Skip counting | Prime factorization | least | common | multiple | smallest | 12 | shared | both

WATCH OUT!

A common mistake

A common mistake in Least Common Multiple is multiplying the two numbers together instead of listing multiples to find the smallest shared one. For example, for 4 and 6, a student calculates 4 × 6 = 24 and calls that the LCM, but the actual least common multiple is 12 (multiples of 4: 4, 8, 12

LESSON 6.12

Least Common Multiple

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Mark and label each value on the number line.**

Find the LCM of 5 and 6 on the number line



STEPS TO TRY
1. Read the labels: some marks are multiples of 5 and some are multiples of 6. The LCM is the first spot that belongs to BOTH.
2. Skip count by 5s (5, 10, 15, …) and by 6s (6, 12, 18, …) as you place marks, watching for where the two lists collide.
3. Place 5 and 6 first, then 10 and 12, and keep alternating up the line. The mark labeled as being in BOTH lists is your LCM.

SHOW YOUR THINKING

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

What is the LCM of 3 and 5?

(A) 3
(B) 5
(C) 15
(D) 30

STEPS TO TRY

1. LCM is about MULTIPLES (counting UP by each number), not factors. You want the smallest number BOTH lists reach.
2. Skip count by 3s in one row and by 5s in another row, then scan for the first number that appears in both rows.
3. Write the 3s row: 3, 6, 9, 12, … Then start the 5s row: 5, 10, … Stop as soon as one number shows up in both rows.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is the LCM of 6 and 8?

(A) 6
(B) 14
(C) 24
(D) 48

STEPS TO TRY

1. Careful: 6 + 8 is not a multiple of either number. You need a number that both 6 AND 8 divide into evenly.
2. List multiples of the LARGER number, 8, and check each one: is it also divisible by 6? Stop at the first hit.
3. Test 8: divisible by 6? No. Test 16: divisible by 6? No. Keep climbing the 8s list until a number divides evenly by 6.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is the LCM of 4 and 10?

(A) 2
(B) 10
(C) 20
(D) 40

STEPS TO TRY

1. The LCM is the smallest number both 4 and 10 count into — it is often much smaller than 4 × 10 = 40.
2. List the multiples of 10 (the bigger number) and test each one for divisibility by 4.
3. Is 10 divisible by 4? 10 ÷ 4 = 2.5, so no. Move up to the next multiple of 10 and run the same test.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

What is the least common multiple of 4 and 6? How did sorting the multiples help you find it?

SENTENCE STARTER The LCM of 4 and 6 is ___ because it is the smallest number in the ___ group.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help